

Supplementary Material

Causal graph recommendation under exposure bias: removing the repeated-walk bias of doubly robust propagation

Theorem, table, section and equation numbers without the prefix S refer to the main text.

S1. Proofs

Throughout, expectations are conditional on Y , and “walk” means a three-hop walk $u - j - v - i$ unless stated otherwise. Its edges are (u, j) , (v, j) and (v, i) .

Equations (3) and (4). Split the plug-in sum $(\tilde{W}\tilde{W}^\top\tilde{W})_{ui} = \sum_{j,v} \tilde{W}_{uj}\tilde{W}_{vj}\tilde{W}_{vi}$ by walk type (Figure 1): (a) $v \neq u, j \neq i$; (b) $v = u, j \neq i$, giving $\sum_{j \neq i} \tilde{W}_{uj}^2 \tilde{W}_{ui}$; (c) $v \neq u, j = i$, giving $\sum_{v \neq u} \tilde{W}_{ui} \tilde{W}_{vi}^2$; (d) $v = u, j = i$, giving $\tilde{W}_{ui}^3$. The corrected estimator keeps (a) and replaces (b)–(d) by $\sum_{j \neq i} C_{uj}^2 W_{uj} \tilde{W}_{ui}$, $\sum_{v \neq u} \tilde{W}_{ui} C_{vi}^2 W_{vi}$ and $C_{ui}^3 W_{ui}$. The difference between plug-in and corrected sums is $\tilde{W}_{ui} \sum_{j \neq i} \Delta_{uj} + \tilde{W}_{ui} \sum_{v \neq u} \Delta_{vi} + (\tilde{W}_{ui}^3 - C_{ui}^3 W_{ui}) = \tilde{W}_{ui}(r_u - \Delta_{ui}) + \tilde{W}_{ui}(\kappa_i - \Delta_{ui}) + (\tilde{W}_{ui}^3 - C_{ui}^3 W_{ui})$, which is Eq. (3). For Eq. (4), $(\tilde{w}_u G)_i = \sum_{j \neq i} \tilde{W}_{uj}(\tilde{W}^\top \tilde{W})_{ji} + \tilde{W}_{ui} \sum_v C_{vi}^2 W_{vi} = (\tilde{W}\tilde{W}^\top\tilde{W})_{ui} - \tilde{W}_{ui}\kappa_i + \tilde{W}_{ui} \sum_v C_{vi}^2 W_{vi}$, and substituting into Eq. (3) cancels the κ_i terms. $\square$

Theorem 1. By the decomposition above, the corrected estimator is a sum over walks of $\prod_e C_e^{k_e} W_e$, the product running over the *distinct* edges of the walk with multiplicities k_e . Under Assumptions 1–2 the W_e of distinct edges

are independent, so the expectation factorises, with $\mathbb{E}W_e = \hat{Y}_e + p_e(Y_e - \hat{Y}_e)/\bar{p}_e = Y_e + \xi_e$. The target walk monomial is $\prod_e C_e^{k_e} Y_e^{k_e} = \prod_e C_e^{k_e} Y_e$ because $Y_e \in \{0, 1\}$. The one-hop term is unbiased edge by edge. $\square$

Corollary 2.. Given $O_{ui} = 0$, $W_{ui} = \hat{Y}_{ui}$ is deterministic and the other edge estimates keep their distribution (Assumption 1). The edge (u, i) belongs to the one-hop term and to exactly the walks of types (b), (c) and (d). In each of them it contributes $C_{ui}^k \hat{Y}_{ui}$ instead of $C_{ui}^k Y_{ui}$, and every other distinct edge contributes its mean Y_e . After the idempotent reduction, F_{ui}^* is affine in Y_{ui} , so $\mathbb{E}[s_{ui} \mid O_{ui} = 0] = F_{ui}^*|_{Y_{ui} \leftarrow \hat{Y}_{ui}} = F_{ui}^* - (Y_{ui} - \hat{Y}_{ui}) \partial F_{ui}^* / \partial Y_{ui}$. The derivative collects aC_{ui} from the one-hop term, $bC_{ui} \sum_{j \neq i} C_{uj}^2 Y_{uj}$ from type (b), $bC_{ui} \sum_{v \neq u} C_{vi}^2 Y_{vi}$ from type (c) and bC_{ui}^3 from type (d). For IPS edges, $W_{ui} = 0$ given $O_{ui} = 0$. $\square$

Theorem 3.. Fix u and i . For an assignment φ of values to the free variables, let $\pi(\varphi)$ be its equality pattern. Two walk edges (U_s, J_t) and $(U_{s'}, J_{t'})$ coincide if and only if $U_s, U_{s'}$ share a user block and $J_t, J_{t'}$ share an item block, so the distinct edges and their multiplicities k_e depend on $\pi(\varphi)$ only. For a pattern π , let $F_\pi(\varphi) = \prod_\epsilon X_{\varphi(\epsilon)}^{(k_\epsilon)}$, the product over the distinct edges ϵ of π , with $X^{(k)} = C^{\odot k} \odot W$ (corrected) or $X^{(k)} = \tilde{W}^{\odot k}$ (plug-in). The plug-in K -hop product equals $\sum_\pi S_\pi^{\text{plug}}$ and the corrected estimator is $\sum_\pi S_\pi^{\text{corr}}$, where $S_\pi = \sum_{\varphi: \pi(\varphi) = \pi} F_\pi(\varphi)$. The two coincide for patterns without a repeated edge, which gives the form of $\hat{T}^{(K)}$ once each S_π is written through unconstrained sums. For $\sigma \geq \pi$ let $g_\pi(\sigma) = \sum_{\varphi: \pi(\varphi) \geq \sigma} F_\pi(\varphi)$: variables in the same block of σ are equal and values of different blocks are unrestricted, so $g_\pi(\sigma)$ is a single tensor contraction over the blocks of σ (with the blocks of U_0 and J_{r+1} fixed to u and i). Grouping the terms of $g_\pi(\sigma)$ by their exact pattern gives

$g_\pi(\sigma) = \sum_{\tau \geq \sigma} f_\pi(\tau)$ with $f_\pi(\tau) = \sum_{\varphi: \pi(\varphi) = \tau} F_\pi(\varphi)$. Möbius inversion on the interval $[\pi, \hat{1}]$ of the product of the two partition lattices (Rota, 1964) yields $S_\pi = f_\pi(\pi) = \sum_{\sigma \geq \pi} \mu(\pi, \sigma) g_\pi(\sigma)$, where $\mu(\pi, \sigma) = \prod_B (-1)^{b_B-1} (b_B - 1)!$ over the blocks B of σ in both lattices and b_B is the number of blocks of π merged into B . Finally, for every φ with $\pi(\varphi) = \pi$ the distinct edges are independent, so $\mathbb{E} F_\pi^{\text{corr}}(\varphi) = \prod_\epsilon C_{\varphi(\epsilon)}^{k_\epsilon} Y_{\varphi(\epsilon)}$, which is the corresponding monomial of $(\tilde{Y} \tilde{Y}^\top)^r \tilde{Y}$ since Y is binary. The number of contraction terms is the number of pairs (π, σ) with $\sigma \geq \pi$ and π containing a repeated edge, which is 5, 99 and 2,790 for $K = 3, 5, 7$. $\square$

Proposition 4.. With correct propensities $\mathbb{E} W_e = Y_e$, and for distinct e, f in the same row $\text{Cov}(W_e, W_f) = (Y_e - \hat{Y}_e)(Y_f - \hat{Y}_f) \text{Cov}(O_e, O_f)/(p_e p_f)$, so $|\text{Cov}(W_e, W_f)| \leq \rho \varepsilon_e \varepsilon_f$. In a type (a) walk the edge (u, j) lies in row u and the edges $(v, j), (v, i)$ in row $v \neq u$; rows are independent, so $\mathbb{E}[W_{uj} W_{vj} W_{vi}] = Y_{uj}(Y_{vj} Y_{vi} + \text{Cov}(W_{vj}, W_{vi}))$ and the bias of the walk is at most $C_{uj} C_{vj} C_{vi} \rho \varepsilon_{vj} \varepsilon_{vi}$. A corrected type (b) walk contributes $C_{uj}^2 C_{ui} W_{uj} W_{ui}$ with two edges in row u , with bias at most $C_{uj}^2 C_{ui} \rho \varepsilon_{uj} \varepsilon_{ui}$. Type (c) walks have their two distinct edges in different rows and type (d) walks a single edge, so they are unbiased. Summing gives the bound. For k of n items drawn uniformly, $\Pr(O_e = O_f = 1) = k(k-1)/(n(n-1))$, so $|\text{Cov}(O_e, O_f)|/(p_e p_f) = (n-k)/(k(n-1)) \leq 1/k$. $\square$

Theorem 5.. By the decomposition of Eq. (3), the plug-in minus the corrected operator is $\tilde{W}_{ui} \sum_{j \neq i} \Delta_{uj} + \tilde{W}_{ui} \sum_{v \neq u} \Delta_{vi} + (\tilde{W}_{ui}^3 - C_{ui}^3 W_{ui})$. Given $O_{ui} = 0$, $\tilde{W}_{ui} = C_{ui} \hat{Y}_{ui}$ is deterministic and the other Δ_e keep their distribution. With correct propensities and binary Y , $\mathbb{E} \Delta_e = C_e^2 (\mathbb{E} W_e^2 - \mathbb{E} W_e) = C_e^2 (\text{Var } W_e + Y_e^2 - Y_e) = C_e^2 V_e$ with $V_e = \frac{1-p_e}{p_e} \varepsilon_e^2$, which gives the stated bias, the last term

being $C_{ui}^3(\hat{Y}_{ui}^3 - \hat{Y}_{ui})$. Part (a) uses $(1-p)/p \geq 1/\max_v p_{vi} - 1$. For (b), with $\mu_e = \mathbb{E}W_e \in [0, 1]$, $\mathbb{E}W_e^2 - \mathbb{E}W_e = \text{Var } W_e - \mu_e(1 - \mu_e) \leq \text{Var } W_e = p_e(1 - p_e)\varepsilon_e^2/\bar{p}_e^2$. If $p_e \geq \tau$ this is $\frac{1-p_e}{p_e}\varepsilon_e^2 \leq (1/\tau - 1)\varepsilon_e^2$; if $p_e < \tau \leq 1/2$ it is $p_e(1 - p_e)\varepsilon_e^2/\tau^2 < \tau(1 - \tau)\varepsilon_e^2/\tau^2$ because $p(1 - p)$ increases on $[0, 1/2]$. At $p_e = \tau$, $\mu_e = Y_e$ and the bound is attained; setting all $p_{vi} = \tau$ gives a bias $C_{ui}\hat{Y}_{ui}(1/\tau - 1)\sum_{v \neq u} C_{vi}^2\varepsilon_{vi}^2$, so the worst case is $\Theta(1/\tau)$. For (c), IPS gives $\tilde{W}_{ui} = 0$ and $C_{ui}^3W_{ui} = 0$ given $O_{ui} = 0$. $\square$

Statement of Theorem S1

Theorem S1 (Variance, clipping and concentration). *Let $\Lambda_e(u, i)$ be the C -weighted mass of the walks from u to i through edge e , and $B = \max(1, 1/\tau)$. Then (a) $\text{Var}(W_e) \leq \varepsilon_e^2/(4\tau^2)$, and $\text{Var}(W_e) \leq \varepsilon_e^2/\tau$ if $\hat{p} = p$; (b) if $\hat{p} = p$, $\text{Var}(\hat{T}_{ui}) \leq \frac{(1+1/\tau)^2}{\tau} \sum_e \varepsilon_e^2 \Lambda_e(u, i)^2$; (c) if $\hat{p} = p$, so that clipping is the only source of bias, $|\mathbb{E}\hat{T}_{ui} - T_{ui}^*| \leq 3\bar{\xi}(1 + \bar{\xi})^2 \sum_{j,v} C_{uj}C_{vj}C_{vi}$ with $\bar{\xi} = \max_e(1 - p_e/\tau)_+ \varepsilon_e$; (d) for any $\hat{p}$, $\Pr(|\hat{T}_{ui} - \mathbb{E}\hat{T}_{ui}| \geq t) \leq 2\exp(-2t^2/\sum_e c_e^2)$ with $c_e = \varepsilon_e B^2 \Lambda_e(u, i)/\tau$.*

Theorem S1.. (a) $\text{Var}(W_e) = p_e(1 - p_e)\varepsilon_e^2/\bar{p}_e^2 \leq \varepsilon_e^2/(4\tau^2)$ for any $\hat{p}$, since $p(1 - p) \leq 1/4$ and $\bar{p}_e \geq \tau$. If $\hat{p} = p$, then $\bar{p} = \max(p, \tau)$: if $p_e \geq \tau$ the variance is at most $(1/\tau - 1)\varepsilon_e^2$, otherwise at most $p_e\varepsilon_e^2/\tau^2 < \varepsilon_e^2/\tau$. Without $\hat{p} = p$ the bound ε_e^2/τ can fail (e.g., $p = 0.5$, $\hat{p} = 0.01$, $\tau = 0.05$ gives $100\varepsilon_e^2 > 20\varepsilon_e^2$). (b) $\hat{T}_{ui}$ is multilinear in the independent W_e and hence affine in each of them. The Efron–Stein inequality (Efron and Stein, 1981) gives $\text{Var}(\hat{T}) \leq \sum_e \mathbb{E}[(\partial_e \hat{T})^2] \text{Var}(W_e)$. The derivative $\partial_e \hat{T}$ is a sum over the walks through e of C -weights times at most two other independent W_f , and with $\hat{p} = p$, $\mu_f = Y_f \in [0, 1]$ and $\mathbb{E}W_f^2 = \mu_f^2 + \text{Var } W_f \leq 1 + 1/\tau$ by (a). Minkowski's

inequality bounds $\|\partial_e \hat{T}\|_2 \leq (1 + 1/\tau)\Lambda_e$. (c) By Theorem 1, the bias is at most $\sum_{\text{walks}} |\kappa_{\text{walk}}| [\prod_f (Y_f + |\xi_f|) - \prod_f Y_f] \leq ((1 + \bar{\xi})^3 - 1) \sum_{j,v} C_{uj} C_{vj} C_{vi}$, and $(1+x)^3 - 1 \leq 3x(1+x)^2$. With $\hat{p} = p$, $|\xi_e| = (1 - p_e/\tau)_+ \varepsilon_e$. (d) Changing O_e moves W_e by at most ε_e/τ . Every edge estimate satisfies $|W_f| \leq B$, so $|\partial_e \hat{T}| \leq B^2 \Lambda_e$, and McDiarmid’s inequality (McDiarmid, 1989) applies. $\square$

Theorem 6. By Definition 1 the estimator uses the post-intervention propensities, and if $\pi p \geq \tau$ the clip does not bind, so they remain correct. DRUP: by Theorem 1, $\mathbb{E}[s] = F^*$, and by Corollary 2, $\mathbb{E}[s \mid O_{ui} = 0]$ is F^* with Y_{ui} replaced by the fixed $\hat{Y}_{ui}$; neither depends on π . Logged graph with fixed C : $\mathbb{E}[O_e Y_e] = p_e Y_e$, and every walk that ends at i contains $k \geq 1$ edges incident to i , so its expectation scales as π^k and η_i is an average of such k . Uncorrected DR: by Theorem 5, $\mathbb{E}[s_{ui} \mid O_{ui} = 0]$ equals the DRUP target plus $C_{ui} \hat{Y}_{ui} (R_u^{(i)} + K_i(\pi))$ minus a term free of π ; $R_u^{(i)}$ involves no edge of item i , and $K_i(\pi) = \sum_{v \neq u} C_{vi}^2 \frac{1 - \pi p_{vi}}{\pi p_{vi}} \varepsilon_{vi}^2$ is strictly decreasing in π . $\square$

S2. Experimental details

Training-free operators. All operators share the score $s = s_1/c_1 + \beta s_3/c_3$ (Section 5) with $\beta \in \{0, 0.03, 0.1, 0.3, 1, 3, 10\}$ or the three-hop term alone; the five-hop variants add $\gamma s_5/c_5$ with $\gamma \in \{0.03, 0.1, 0.3, 1, 3\}$. The grids are: normalisation exponent $\alpha \in \{0.3, 0.5, 0.7\}$ (Coat), $\{0.5, 0.7\}$ (Yahoo!R3, KuaiRand) and $\{0.3, 0.5\}$ (KuaiRec); clip $\tau \in \{0.01, 0.02, 0.05, 0.1, 0.2\}$ (Coat), $\{0.1, 0.2\}$ (Yahoo!R3), $\{0.02, 0.05, 0.1\}$ (KuaiRand) and $\{0.05, 0.1, 0.2\}$ (KuaiRec); imputation ridge $\lambda \in \{1, 5, 20\}$ (Coat), $\{20\}$ (Yahoo!R3, KuaiRand) and $\{5, 20\}$ (KuaiRec); additive or low-rank (rank 32) imputation; control-variate weight $\in \{0, 0.25, 0.5, 1\}$ on

Yahoo!R3, $\{0, 0.5, 1\}$ on KuaiRand and 1 elsewhere; and degree source $\in \{\hat{Y}, \text{cross-fitted } W\}$ for the debiased operators. The degree source is a searched hyper-parameter. $\hat{Y}$ -degrees satisfy Assumption 2 whenever $\hat{Y}$ does; cross-fitted W -degrees do not, since the weight at a pair then depends on the exposures of other pairs, so configurations with W -degrees (selected on Yahoo!R3) lie outside the theorems. Only $\hat{Y}$ -degrees make the certificates of Theorem S4 informative. EASE uses $\lambda \in \{10, 100, 1000\}$; GF-CF uses four settings of (ideal low-pass weight, singular vectors): $(0, 0)$, $(0.3, 64)$, $(0.3, 256)$ and $(1, 256)$. BSPM (Choi et al., 2023) uses blurring time $\in \{1, 2\}$, sharpening time $\in \{0, 0.1, 0.3\}$ and ideal low-pass weight $\in \{0, 0.3\}$ with one Euler step each. On the DR graph, EASE, GF-CF and BSPM search the same nuisance grid as DR adjacency. The sample-split variants are run for five draws of the split-off pairs; each draw is tuned separately, and the tables report the mean over draws. The propensity model is the one shipped with Coat and a logistic exposure model on user and item exposure rates elsewhere; on Yahoo!R3 it was selected on validation data against an outcome-dependent Naive-Bayes model. All nuisances are cross-fitted over ten random folds of pairs, except for the sample-split variants, whose imputation (and propensity model, where it is estimated) is fitted on a random 20% of the pairs.

Trained models. The grid combines the learning rates 10^{-2} , $3 \cdot 10^{-3}$ and 10^{-3} with the weight decays 10^{-5} , 10^{-4} and (Coat, and the MF models on Yahoo!R3) 10^{-3} . Embeddings have 64 dimensions (Coat: 32 or 64). The graph models with the widest search (LightGCN and r-AdjNorm on Yahoo!R3, the pointwise and BPR LightGCN on KuaiRec, all graph models

on Coat) also search one to three layers and 64 or 128 dimensions (Coat: 32 or 64); the others use two layers. PDA searches $\gamma \in \{0.1, 0.2\}$, r-AdjNorm $r \in \{0.3, 0.7\}$, SimGCL the contrastive weight $\in \{0.1, 0.5\}$ (noise 0.1, temperature 0.2, InfoNCE over at most 2,048 users and items of a batch) and MACR $c \in \{0.1, 0.3\}$; iALS searches the confidence $\alpha \in \{3, 10, 30\}$ and the ridge $\in \{1, 10, 100, 1000\}$. DR-JL and MRDR learn an MF imputation model jointly with the prediction model (Wang et al., 2019; Guo et al., 2021). BPR models draw one uniformly sampled negative per positive. Batch sizes are 256 (Coat), 1,024 (Yahoo! R3), 8,192 (KuaiRand) and 32,768 (KuaiRec, with 2M sampled pairs per epoch for the pointwise losses); training runs for at most 100, 60, 40 and 30 epochs. Because the training log is the same for every split, each configuration is trained once and evaluated after every epoch on the validation part of every split. Each split stops on its own validation score (patience 5 on Coat and Yahoo! R3, 3 otherwise) and selects its own configuration and epoch; training continues until every split has stopped, and a split never uses epochs after its own stopping point, so no split’s selection depends on another split’s validation users, who may be its test users. The split-0 selection is re-trained with three (KuaiRec, KuaiRand: two) further seeds to measure seed variance. We did not run DICE, StableDR and methods that train on the uniform data (CausE, AutoDebias): in our protocol the uniform data serve only for selection and testing, for every method. Table S1 lists the number of configurations each method was selected from.

Evaluation and tests. Coat splits each user’s candidates (30% validation), Yahoo! R3, KuaiRand and KuaiRec split the test users (30% validation), over 10, 5, 5 and 5 random splits. Users without a positive candidate are skipped.

For a pair of methods, each test user’s metric difference is averaged over the splits in which the user is in the test part, and we report the mean difference over users with a 95% t -interval and the paired t -test. The Holm family is every comparison run for a dataset, with all references together (DRUP, DR adjacency, five-hop DRUP and DRUP-split against every other method); the Wilcoxon signed-rank p -values are adjusted over the same family. Within-reference adjustments are in the result files. The Wilcoxon signed-rank test gives the same conclusions except for the five-hop DRUP against DR comparison on KuaiRec ($p = 0.66$ against 0.011) and for DRUP-split against pointwise LightGCN on KuaiRec ($p_{\text{Holm}} = 0.48$ against $3 \cdot 10^{-4}$). The corrected resampled t -test over split means (Nadeau and Bengio, 2003) is far more conservative with five or ten splits and is reported in the result files. All tests condition on the training log and the fitted nuisances.

Hardware and code. All experiments ran on four CPU cores with 16 GB of memory, except the KuaiRand debiased and sample-split operators, which ran on a Google Colab T4 GPU (same code and seeds; per-user ranking metrics are computed once per score matrix and match the per-user routine to 10^{-15}). GF-CF uses a randomised SVD, so its scores vary by about 0.002 between runs. Code, configurations and result files are available at the repository given on the title page.

S3. Structural properties of linear propagation

The results of this section use only that, given the nuisances, s is a fixed linear function of the edge estimates. They hold for DRUP and equally for

the uncorrected operators and for linear LightGCN, and they describe the recommender with its nuisances held fixed.

Proposition S2 (Affine in the user’s own data). *Fix u and a candidate i with $O_{ui} = 0$, and let $G^{(-u)}$ be G without u ’s own contribution. With frozen nuisances, $s_{ui} = \text{const} + \sum_{j \neq i} \omega_j w_{uj}$ with $\omega_j = b(c_{uj}G_{ji}^{(-u)} + c_{ui}w_{ui}c_{uj}^2)$. Hence un-logging interaction j , i.e., replacing w_{uj} by its unlogged value $\hat{Y}_{uj}$, changes s_{ui} by exactly $\phi_j = \omega_j(w_{uj} - \hat{Y}_{uj})$, removal effects add up over any subset, and completeness holds exactly.*

Because the removal game is additive, the ϕ_j are also its Shapley values; this is a consequence of linearity, and the attribution is relative to the stated baseline (Kumar et al., 2020).

Theorem S3 (Optimal minimal counterfactual explanation). *For candidates i, k with $s_{ui} > s_{uk}$, sorting $g_j = \phi_j^{(i)} - \phi_j^{(k)}$ in decreasing order and removing interactions until the cumulative sum exceeds the margin yields a minimum-cardinality set whose removal makes k outrank i . If the positive g_j do not sum above the margin, no such set exists. The cost is $O(n + d_u \log d_u)$.*

Theorem S4 (Certificates against injected profiles, frozen nuisances). *With frozen nuisances, one injected user v changes u ’s three-hop score of item k by $\Delta_k = x_k(D + a_k c_{vk}) - a_k x_k^2$, where $a = \tilde{w}_u$, $x = c_v \odot w_v$ and $D = \langle a, x \rangle$. If v logs at most L interactions with arbitrary items, labels and propensities $\geq \tau$, then D and x_k lie in closed-form intervals. Δ_k is linear in D and quadratic in x_k , so its extremes $[\underline{\Delta}_k, \overline{\Delta}_k]$ over the box formed by the two intervals are attained at its edges or at the vertex of the quadratic. The box contains every feasible (D, x_k) but ignores that both are determined by*

the same L interactions, so $[\underline{\Delta}_k, \overline{\Delta}_k]$ contains the feasible range of Δ_k and is conservative rather than exact. Item t provably stays out of u 's top- K against any F such users if $s_{ut} + F\overline{\Delta}_t$ is below the $(K+1)$ -th largest value of $s_{uk} + F\underline{\Delta}_k$ over the other candidates. With unclipped IPS, the influence of a single interaction is unbounded.

The same argument certifies the uncorrected operators (with $\Delta_k = x_k D$). Degrees taken from $\hat{Y}$ make the fake user's weight c_v known; degrees taken from W would be attacker-controlled.

Theorem S5 (Joint differential privacy with public nuisances). *Let the population-level parts of the nuisances (the propensity model, the item parameters of $\hat{Y}$, the item degrees in C and the radius R), the hyper-parameters and the constants a, b of the score be fixed a priori or computed from public data and from the release, and every user-level part from the user's own row. Scale each private row so that $\|\tilde{w}_v\|_2 \leq R$ and $\|c_v^{\odot 2} \odot w_v\|_2 \leq R^2$. Releasing G , computed over the private rows, plus symmetric Gaussian noise of scale $\sigma = \sigma_{\text{aG}}(\epsilon, \delta; 2R^2)$, the analytic Gaussian calibration (Balle and Wang, 2018), is (ϵ, δ) -DP under adding or removing one user, for every $\epsilon > 0$. Scoring by (4) uses only the release and the user's own row, so the recommender is (ϵ, δ) -jointly DP (Hsu et al., 2014). Post-processing such as low-rank denoising is free.*

Row scaling changes the estimate for users whose rows exceed R , so Theorems 1 and 6 hold for the private variant only up to this shrinkage.

Theorem S6 (Exposure-capped allocation). *Let N users each receive K of their candidates and let item i be recommended to at most cap_i users. Let $\hat{x}$*

maximise $\langle x, s \rangle$ over such allocations. The constraint matrix is totally unimodular, so the linear relaxation has an integral optimum, and min-cost flow on the scores rounded to a grid of L levels finds $\hat{x}$ exactly up to the rounding gap $\Gamma \leq NK(\max s - \min s)/L$. All caps hold, at least $NK/\max_i \text{cap}_i$ items are recommended, and for the full-exposure utility $U(x) = \langle x, F^* \rangle$,

$$U(\hat{x}) \geq \max_x U(x) - \Gamma - 2 \max_x |\langle x, s - F^* \rangle|.$$

If $\mathbb{E}s = F^*$ (DRUP under Assumption 2 with $\xi \equiv 0$), McDiarmid's inequality and a union bound over all mn pairs give $\max_x |\langle x, s - F^* \rangle| \leq NK t_\vartheta$ with probability $1 - \vartheta$, where

$$t_\vartheta = \max_{u,i} \left(\frac{1}{2} \sum_e \bar{c}_e(u, i)^2 \ln(2mn/\vartheta) \right)^{1/2},$$

$$\bar{c}_e(u, i) = b c_e(u, i) + a C_{ui} \varepsilon_{ui} \mathbb{K}[e = (u, i)]/\tau,$$

$c_e(u, i)$ is the constant of Theorem S1(d) for the three-hop term and the second part is the sensitivity of the one-hop term. The union runs over all pairs because the candidate set depends on the log. For logged-graph propagation, $\mathbb{E}s = F^*(P \odot Y)$, and the last term does not vanish.

S3.1. Proofs

Proposition S2 and Theorem S3. With frozen nuisances, substitute $G = G^{(-u)} + \tilde{w}_u^\top \tilde{w}_u - \text{diag}(\tilde{w}_u^2) + \text{diag}(c_u^2 w_u)$ into (4). The terms $\tilde{w}_{ui} \sum_j \tilde{w}_{uj}^2$ cancel against the local correction, leaving $\tilde{w}_{ui} \sum_j c_{uj}^2 w_{uj}$, and every remaining term involving w_{uj} ($j \neq i$) is linear, with coefficient $c_{uj} G_{ji}^{(-u)} + c_{ui} w_{ui} c_{uj}^2$ in $\hat{T}_{ui}$. Since $O_{ui} = 0$, the one-hop term $a\tilde{W}_{ui}$ does not involve w_{uj} , so the coefficient in $s_{ui} = a\tilde{W}_{ui} + b\hat{T}_{ui}$ is ω_j . Un-logging j replaces w_{uj} by $\hat{Y}_{uj}$, which changes s_{ui}

by $\omega_j(w_{uj} - \hat{Y}_{uj})$; the game is additive, so its Shapley values are the individual effects. For Theorem S3, the margin after removing S is $s_{ui} - s_{uk} - \sum_{j \in S} g_j$, so minimising $|S|$ under $\sum_S g_j > \text{margin}$ selects the largest g_j . $\square$

Theorem S4.. Adding row v adds contrib_v to G and leaves u 's local terms unchanged, so $\Delta = \tilde{w}_u \text{contrib}_v = (a \cdot x)x - a \odot x^2 + a \odot c_v \odot x$. A logged DR entry satisfies $w \in [\hat{Y} - \hat{Y}/\tau, \hat{Y} + (1 - \hat{Y})/\tau]$ for any label and any $p \geq \tau$, and an unlogged one equals $\hat{Y}$. D is maximised (minimised) by assigning the L slots to the largest positive (negative) deviations, and x_k ranges over the values of a logged or unlogged entry. Every feasible profile gives a point (D, x_k) of the product of the two intervals, so the extremes of Δ_k over this box bound its feasible extremes; they are not attained in general, because the box treats D and x_k as free although both depend on the same at most L entries. For fixed x_k the effect is linear in D , so its extremes over the box are at the ends of the D interval; for fixed D it is quadratic in x_k , with extremes at the ends or at the clamped vertex. With frozen nuisances G is a sum of per-user contributions, so effects add over fake users. For unclipped IPS, $|W_e| = 1/p_e$ is unbounded. $\square$

Theorem S5.. Because the population-level nuisances are public and every user-level part depends only on the user's own row, adding or removing user v changes G by contrib_v alone. With the row scaling, $\|\text{contrib}_v\|_F \leq \|\tilde{w}_v \tilde{w}_v^\top - \text{diag}(\tilde{w}_v^2)\|_F + \|c_v^2 w_v\|_2 \leq R^2 + R^2$, so the ℓ_2 -sensitivity of the upper triangle is at most $2R^2$. The analytic Gaussian mechanism is (ϵ, δ) -DP for every $\epsilon > 0$ exactly when $\Phi(\frac{\Delta}{2\sigma} - \frac{\epsilon\sigma}{\Delta}) - e^\epsilon \Phi(-\frac{\Delta}{2\sigma} - \frac{\epsilon\sigma}{\Delta}) \leq \delta$ (Theorem 8 of Balle and Wang (2018)); we use the smallest such σ . The classical calibration $\sigma = \Delta \sqrt{2 \ln(1.25/\delta)}/\epsilon$ (Dwork and Roth, 2014) is only valid for $\epsilon < 1$. Each

user's output is a function of the release and the user's own data, which gives joint DP by the billboard lemma (Hsu et al., 2014). $\square$

Theorem S6. The allocation polytope is defined by the incidence matrix of the bipartite user–candidate graph with degree bounds, which is totally unimodular, so the linear relaxation has an integral optimal vertex and min-cost flow finds it. Rounding changes each score by at most $h/2$ with $h = (\max s - \min s)/L$. If x^* is optimal for s and $\hat{x}$ for the rounded scores $\tilde{s}$, then $\langle x^*, s \rangle \leq \langle x^*, \tilde{s} \rangle + NK h/2 \leq \langle \hat{x}, \tilde{s} \rangle + NK h/2 \leq \langle \hat{x}, s \rangle + NK h$, so $\Gamma \leq NK h$. Each item is recommended at most $\max_i \text{cap}_i$ times and NK slots are filled, which gives the coverage bound. With x_F optimal for U , $U(\hat{x}) \geq \langle \hat{x}, s \rangle - |\langle \hat{x}, s - F^* \rangle| \geq \langle x_F, s \rangle - \Gamma - |\langle \hat{x}, s - F^* \rangle| \geq U(x_F) - \Gamma - 2 \max_x |\langle x, s - F^* \rangle|$. For the concentration step, $s_{ui} = a\tilde{W}_{ui} + b\hat{T}_{ui}$ is a function of the independent exposures. Changing O_e moves $b\hat{T}_{ui}$ by at most $b c_e(u, i)$ (proof of Theorem S1(d)) and, for $e = (u, i)$, moves $a\tilde{W}_{ui}$ by at most $a C_{ui} \varepsilon_{ui} / \tau$, so s_{ui} has bounded differences $\bar{c}_e(u, i)$. If $\mathbb{E}s = F^*$, McDiarmid's inequality gives $\Pr(|s_{ui} - F_{ui}^*| \geq t) \leq 2 \exp(-2t^2 / \sum_e \bar{c}_e(u, i)^2)$. The candidate set depends on O , so we take a union bound over all mn pairs: with probability $1 - \delta$, $|s_{ui} - F_{ui}^*| \leq t_\delta$ for every pair, in particular for every candidate, and an allocation sums NK such terms. $\square$

S4. Structural properties on real data

The properties of Section S3 hold for the model with frozen (explanations, certificates) or public (privacy) nuisances, and for any linear operator; we check them for DRUP and report where they are informative.

Explanations. Attributions recomputed against full re-propagation are exact to floating-point precision (relative error 10^{-16} in float64 on Coat and Yahoo!R3, $3 \cdot 10^{-6}$ in float32 on KuaiRec; Table S3), and removing the interactions with the largest attribution moves the top item further down than removing random or similar ones, so the attributions are faithful. The counterfactual analysis mostly certifies that no explanation from the user’s own data exists: on Coat 94% of the top recommendations cannot be overturned by removing any subset of the user’s own interactions, on KuaiRec 52%, and on Yahoo!R3 10%; the minimal explanation otherwise has a median size of 2, 7 and 1 interactions. Such recommendations rest on population-level evidence (through G), which the attribution to a user’s own interactions does not explain.

Robustness. No certificate was violated in any run (Table S4). With imputation-based degrees, DR adjacency and DRUP certify every Coat user against one or two fake profiles and 93–95% of users on Coat and KuaiRec against five, and realised attacks with up to 100 fake users placed the target in a KuaiRec top-20 for at most 0.3% of users under IPS, DR adjacency and DRUP (2.7% for the logged graph). Certificates are informative only for small budgets: at $F = 20$ they cover 10–26% of users, and at 200 fake users the target reaches 39% of KuaiRec users under DR and DRUP, more than under IPS (10%) and the logged graph (29%). On Yahoo!R3, where DRUP selects IPS edges with cross-fitted W -degrees, the fake users’ degrees are attacker-controlled and no user is certified. When the nuisances are re-fitted on the poisoned log, which the certificates do not cover, none of the certified user–target pairs of DRUP (5,196 on Coat, 2,821 on KuaiRec) was violated,

and the realised hit rates were lower than with frozen nuisances (Table S12 in the section on additional results), because the fake users’ own imputation raises their degree and lowers their weight.

Privacy. Table S5 reports the jointly private variant of Theorem S5, in which only public users inform the population-level nuisances and the score constants c_1, c_3 . The configuration is fixed a priori (clip 0.1, ridge 20, additive imputation, $\alpha = 0.5$, control-variate weight 1, $\beta = 1$), not selected on the test data. Useful privacy needs a large user base and a weak guarantee, and the public-nuisance model can itself be the limit. On KuaiRec, with 717 public users, private DRUP reaches nDCG@20 of 0.618 at $\epsilon = 16$ and 0.598 at $\epsilon = 8$, against 0.634 without noise, 0.637 with all nuisances and 0.577 for popularity; it falls below popularity for $\epsilon \leq 4$. On Coat (29 public users) the noise-free model reaches 0.548, only just above popularity (0.527), and it is below popularity at every finite ϵ (0.438 at $\epsilon = 16$). On Yahoo! R3 the noise-free model with public nuisances reaches 0.492 and so is below popularity (0.525) at every ϵ , although the non-private model with all nuisances reaches 0.652: 1,540 public users do not estimate the item parameters of a 2%-dense log. The privacy variant is therefore useful only on the largest log. Privatising the nuisance models with a composed budget is left open.

S4.1. Exposure-capped allocation

Table S6 applies the exposure-capped allocation of Theorem S6 to every operator. The min-cost-flow solver returns the optimum up to its rounding gap (a relative gap below $4 \cdot 10^{-5}$ in every run), all caps hold, and coverage never falls below the guaranteed $NK/\text{max cap}$. On KuaiRec, where exposure

is extremely concentrated, DR adjacency and DRUP are the most accurate operators under every cap, for example 0.199 against 0.173 (logged graph) and 0.178 (IPS) at $c = 5$, and 0.140 against 0.115 and 0.121 at $c = 2$; the two agree to the third decimal under every cap, so the correction does not change the allocation either. Under the same cap, however, their lists remain more concentrated (Gini 0.80 against 0.72 at $c = 5$), and at equal Gini the frontier over each operator’s grid (Table S11) favours the correlation-based operators on Coat. On Coat, whose random test items are already diverse, capping costs DRUP more accuracy than it costs the logged graph; on Yahoo!R3 exposure is not concentrated (Gini 0.27) and caps down to twice the fair share cost every operator at most 0.002. The allocation is thus the mechanism that controls concentration, and the DR graph is the better input to it when exposure in the log is very uneven.

S5. Additional results

S5.1. Does the correction matter on sparser logs?

Theorem 5 predicts a larger bias when propensities are small. We therefore thinned the KuaiRec log to 10% and 3% of its exposures, which lowers the median logged propensity from 0.48 to 0.045 and 0.013, re-fitted the nuisances and re-selected every operator (Table S20). The corrected and the uncorrected operators still rank almost identically: their top-20 lists overlap by at least 93%, the rank correlation is at least 0.999, and the accuracy differences stay below 0.0005 nDCG@20 in either direction, at three and at five hops. On these data the repeated-walk term is small compared with the score differences that decide the ranking, and the value of the correction lies

in the properties of the scores, not in the rankings they induce.

S5.2. Clip and exposure invariance; exposure concentration

Table S22 shows the distributional side. On KuaiRec, the exposure-conditional bias drops from 0.83 (logged graph) and 0.80 (IPS) to 0.33 for DRUP and DR adjacency: given their true quality, items are ranked much less by how often they were shown. At the same time DRUP’s top-20 lists are concentrated on 1.2% of the candidate items, against 15% for the logged graph, because the estimated full-exposure scores are themselves concentrated on few items; on Coat its Gini coefficient is 0.60 against 0.34. Exposure invariance and exposure concentration are different properties, and DRUP improves the first at the expense of the second. The user-group gaps do not differ between operators. They are clearly non-zero on Coat (men 0.11 above women), on Yahoo!R3 (inactive users 0.046 below active ones) and on KuaiRand (inactive users 0.014 above active ones), under every operator, and not on KuaiRec. On KuaiRand the lists of DRUP, IPS and the logged graph are almost equally concentrated (Gini 0.43, coverage 0.98), as the scores of all three are close to each other there.

S5.3. Stakeholders and regulation

Item providers. Exposure invariance means that an item’s expected score does not depend on how often it was observed; it is not equal exposure, and it comes with more concentrated lists (Table S22) that a retrained system could entrench. Deployments should report concentration next to invariance and use an explicit allocation mechanism. *Users.* The removal attributions describe the model with its nuisances frozen and are not an account of data

deletion. *Platforms and auditors*. The clip τ and the control-variate weight λ trade invariance against accuracy; one option is to select the most accurate configuration subject to a bound on the elasticity measured in a thinning audit, whose power is limited. The audit is a first-party instrument: it needs the log, the propensities, the deployed hyper-parameters and the ability to re-run the pipeline. Article 40 of the Digital Services Act (European Parliament and Council of the European Union, 2022) gives vetted researchers access to data of very large online platforms and search engines; whether it extends to re-running a proprietary pipeline is not settled by the text, and we do not claim that it does. The Act’s risk-assessment and audit obligations (Articles 34–37, Delegated Regulation (EU) 2024/436 (European Commission, 2024)) concern compliance of very large providers, not properties of estimators, and our results can at most inform a risk assessment.

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Table S1: Number of configurations searched per method (per split).

Method	Coat	Yahoo! R3	KuaiRand	KuaiRec
Popularity (non-personalised)	1	1	1	1
Imputation only (non-personalised)	6	2	2	4
MF (trained, pointwise)	18	9	6	6
IPS-MF (trained, pointwise)	18	9	–	6
DR-MF (trained, pointwise)	18	9	6	6
DR-JL (trained, pointwise)	18	9	6	6
MRDR (trained, pointwise)	18	9	–	6
iALS (trained, pointwise)	24	12	12	12
LightGCN (trained, pointwise)	54	6	6	36
DR-LightGCN (trained, pointwise)	54	6	–	6
MF (BPR) (trained, pairwise)	18	9	6	6
PDA (trained, pairwise)	36	18	–	12
MACR (trained, pairwise)	36	18	12	12
LightGCN (BPR) (trained, pairwise)	54	36	6	36
r-AdjNorm (trained, pairwise)	108	72	12	12
NAVIP (trained, pairwise)	54	36	6	6
SimGCL (trained, pairwise)	108	12	12	12
linear LightGCN / r-AdjNorm (training-free)	24	16	16	16
EASE (training-free)	3	3	3	3
GF-CF (training-free)	4	4	4	4
BSPM (training-free)	12	12	12	12
IPS adjacency (NAVIP-style) (training-free)	720	64	96	192
IPS + walk correction (training-free)	720	64	96	192
propagation over $\hat{Y}$ only (no residual) (training-free)	1440	128	–	384
EASE on DR graph (training-free)	90	48	54	36
GF-CF on DR graph (training-free)	120	64	72	48
BSPM on DR graph (training-free)	360	192	216	144
DR adjacency (training-free)	1440	512	576	384
DRUP (training-free)	1440	512	576	384
DR adjacency, 5 hops (training-free)	6300	–	–	420
DRUP, 5 hops (training-free)	6300	–	–	420
DR adjacency, sample split (training-free)	720	256	288	192
DRUP-split (training-free)	720	256	288	192

Table S2: Tuning budget. Expected test score of DRUP when its configuration is selected on validation data among k configurations drawn at random from its grid (exact, averaged over splits; Dodge et al. (Dodge et al., 2019)), with k the budget of the most accurate trained model. Configurations: a trained configuration keeps its best epoch per split, and every value of β is a separate DRUP configuration. Evaluations: every (configuration, epoch) of a trained model counts as one validation evaluation.

Data	best trained model	configurations		evaluations			
		k	DRUP	k	DRUP	DRUP, all	trained
Coat	iALS	24	0.5487	295	0.5451	0.5467	0.5812
Yahoo! R3	r-AdjNorm	72	0.6565	1765	0.6567	0.6567	0.6742
KuaiRec	DR-JL	6	0.6182	103	0.6362	0.6378	0.6332

Table S3: Exact explanations (Proposition S2, Theorem S3) on 200 random test users: largest relative error of completeness and subset additivity against re-propagation; rank of the top-1 item after un-logging the five interactions with the largest attribution, five random ones, or the five most similar ones (0 = still first); share of top-1 recommendations for which no subset of the user’s own interactions makes the second item overtake it; median size of the minimal counterfactual explanation otherwise. Every explanation was verified by re-scoring.

Data	max. rel. error	top-1 rank after 5 removals (attr. / random / similarity)	no CF explanation	median CF size
Coat	6.1e-16	0.04 / 0.01 / 0.01	94%	2.0
Yahoo! R3	8.7e-16	1.65 / 0.38 / 1.40	10%	1.0
KuaiRec	3.1e-06	0.03 / 0.01 / 0.01	52%	7.0

Table S4: Injection attack: F fake users log a low-popularity target and the most popular items. Realised hit rate of the target in real users’ top- K , and the fraction of real users for whom Theorem S4 certifies that the target cannot enter the top- K against *any* F profiles of that size (frozen nuisances). No certificate was violated.

Data	Operator	realised hit rate			certified fraction		
Coat	$F =$	1	5	20	1	5	20
	logged graph	0.004	0.009	0.444	0.00	0.00	0.00
	IPS adjacency	0.001	0.269	0.661	0.15	0.00	0.00
	DR adjacency	0.000	0.000	0.029	1.00	0.94	0.11
	DRUP	0.000	0.000	0.028	1.00	0.93	0.10
Yahoo! R3	$F =$	1	5	100	1	5	100
	logged graph	0.065	0.065	0.199	0.12	0.00	0.00
	IPS adjacency	0.065	0.065	0.171	0.00	0.00	0.00
	DR adjacency	0.065	0.065	0.171	0.00	0.00	0.00
	DRUP	0.065	0.065	0.171	0.00	0.00	0.00
KuaiRec	$F =$	1	5	200	1	5	200
	logged graph	0.002	0.002	0.288	0.00	0.00	0.00
	IPS adjacency	0.001	0.001	0.105	0.98	0.61	0.00
	DR adjacency	0.000	0.000	0.393	1.00	0.95	0.00
	DRUP	0.000	0.000	0.393	1.00	0.95	0.00

Table S5: nDCG of jointly private DRUP ($\delta = 10^{-5}$, analytic Gaussian mechanism, denoising rank 32 fixed in advance). Population-level nuisances come from a public 10% of the users (non-test users where available; on Coat, whose users are all test users, the public users are excluded from evaluation); “all nuisances” is the non-private model with all nuisances fitted on the whole log. Other ranks in Table S13.

Data	$\epsilon = 0.5$	1	2	4	8	16	∞	all nuisances	popularity
Coat	0.382	0.385	0.403	0.385	0.401	0.438	0.548	0.555	0.527
Yahoo! R3	0.481	0.490	0.491	0.493	0.492	0.493	0.492	0.652	0.525
KuaiRec	0.174	0.175	0.154	0.412	0.598	0.618	0.634	0.637	0.577

Table S6: Exposure-capped re-ranking (Theorem S6, exact min-cost flow). Each item may be recommended to at most c times its fair share NK/n of users. Cells: nDCG@ K (Gini@ K). Every cap held; the full range of caps and the accuracy–concentration frontier are in the section on additional results.

Data	Operator	no cap	$c = 5$	$c = 2$
Coat	logged graph	0.502 (0.34)	0.501 (0.34)	0.494 (0.32)
	IPS adjacency	0.507 (0.35)	0.503 (0.34)	0.493 (0.33)
	DR adjacency	0.489 (0.60)	0.489 (0.60)	0.475 (0.48)
	DRUP	0.489 (0.60)	0.489 (0.60)	0.476 (0.48)
Yahoo! R3	logged graph	0.658 (0.28)	0.658 (0.28)	0.658 (0.27)
	IPS adjacency	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)
	DR adjacency	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)
	DRUP	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)
KuaiRec	logged graph	0.603 (0.99)	0.173 (0.72)	0.115 (0.44)
	IPS adjacency	0.612 (0.99)	0.178 (0.79)	0.121 (0.49)
	DR adjacency	0.639 (0.99)	0.199 (0.80)	0.140 (0.50)
	DRUP	0.638 (0.99)	0.199 (0.80)	0.140 (0.50)

Table S7: Stress tests outside Assumption 1 and with misspecified propensities (log-normal noise with $\sigma = 0.5$). Relative $|\text{bias}|$ on unexposed candidates.

Scenario ($K = 3$, rel. $ \text{bias} $ on unexposed candidates)	IPS	IPS + WC	DR	DRUP
independent exposures (Assumption 1)	0.029	0.029	1.137	0.019
fixed-size slates per user (Proposition 4)	0.219	0.219	1.119	0.024
misspecified propensities, exact imputation	0.568	0.568	0.000	0.000
misspecified propensities, noisy imputation	0.568	0.568	1.916	0.158
independent exposures (Assumption 1), $K = 5$	4.667	0.059	5.053	0.040
fixed-size slates per user (Proposition 4), $K = 5$	2.392	0.416	4.633	0.032
misspecified propensities, noisy imputation, $K = 5$	19.266	1.162	16.027	0.204

Table S8: Nuisance protocol in the 40-user, 60-item model of Table 4 with true and with estimated (logistic exposure model, misspecified here) propensities, nuisances from an independent log, cross-fitted, or fitted on the same log without cross-fitting, and degrees from $\hat{Y}$ or from edge estimates. Relative $|\text{bias}|$ of the three-hop term and of the full score with data-dependent constants (3,000 replications).

Propensities	Nuisances	degrees	IPS edges			DR edges		
			uncorr.	DRUP	$s, \text{ data } c$	uncorr.	DRUP	$s, \text{ data } c$
true	independent log	$\hat{Y}$	2.15	0.10	0.05	1.90	0.09	0.35
	independent log	W	5.04	0.25	0.10	3.59	0.18	0.35
	cross-fitted	$\hat{Y}$	0.39	0.64	0.15	0.61	0.80	0.62
	cross-fitted	W	3.12	0.81	0.11	4.11	0.73	0.49
	same log	$\hat{Y}$	0.42	0.70	0.18	0.59	0.70	0.33
	same log	W	0.94	0.97	0.49	0.80	0.86	0.37
estimated	independent log	$\hat{Y}$	29.57	3.70	0.15	25.57	2.11	0.53
	independent log	W	92.66	7.44	0.15	47.74	3.00	0.50
	cross-fitted	$\hat{Y}$	1.29	0.48	0.11	3.41	1.23	0.92
	cross-fitted	W	83.73	0.49	0.18	27.63	0.73	0.59
	same log	$\hat{Y}$	0.60	0.82	0.21	0.64	0.72	0.33
	same log	W	0.93	0.97	0.53	0.83	0.88	0.38

Table S9: Tightness of Theorem S1 in a 12-user, 15-item model (20,000 replications, all propensities $\geq \tau$), over all (u, i) .

	$\tau = 0.3$	$\tau = 0.1$
variance bound (b) / Monte-Carlo variance, median	1582	8422
variance bound (b) / Monte-Carlo variance, minimum	471	2126
McDiarmid half-width (d) / empirical 95% quantile, median	131	1831
McDiarmid half-width (d) / empirical 95% quantile, minimum	73	685

Table S10: Exposure-capped re-ranking for all caps. Cells: nDCG@ K (Gini@ K).

Coat	none	$c = 20$	$c = 10$	$c = 5$	$c = 3$	$c = 2$	$c = 1.5$
logged graph	0.502 (0.34)	0.494 (0.34)	0.501 (0.34)	0.501 (0.34)	0.493 (0.33)	0.494 (0.32)	0.487 (0.28)
IPS adjacency	0.507 (0.35)	0.497 (0.35)	0.506 (0.35)	0.503 (0.34)	0.504 (0.36)	0.493 (0.33)	0.486 (0.29)
DR adjacency	0.489 (0.60)	0.489 (0.60)	0.489 (0.60)	0.489 (0.60)	0.485 (0.58)	0.475 (0.48)	0.444 (0.37)
DRUP	0.489 (0.60)	0.489 (0.60)	0.489 (0.60)	0.489 (0.60)	0.485 (0.58)	0.476 (0.48)	0.443 (0.37)
KuaiRec	none	$c = 20$	$c = 10$	$c = 5$	$c = 3$	$c = 2$	$c = 1.5$
logged graph	0.603 (0.99)	0.300 (0.91)	0.228 (0.84)	0.173 (0.72)	0.139 (0.60)	0.115 (0.44)	0.102 (0.31)
IPS adjacency	0.612 (0.99)	0.307 (0.94)	0.234 (0.89)	0.178 (0.79)	0.146 (0.66)	0.121 (0.49)	0.107 (0.34)
DR adjacency	0.639 (0.99)	0.335 (0.95)	0.257 (0.90)	0.199 (0.80)	0.165 (0.67)	0.140 (0.50)	0.125 (0.35)
DRUP	0.638 (0.99)	0.335 (0.95)	0.256 (0.90)	0.199 (0.80)	0.164 (0.67)	0.140 (0.50)	0.125 (0.35)
Yahoo! R3	none	$c = 20$	$c = 10$	$c = 5$	$c = 3$	$c = 2$	$c = 1.5$
logged graph	0.658 (0.28)	0.659 (0.28)	0.660 (0.28)	0.658 (0.28)	0.656 (0.28)	0.658 (0.27)	0.652 (0.23)
IPS adjacency	0.656 (0.27)	0.657 (0.27)	0.657 (0.27)	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)	0.651 (0.22)
DR adjacency	0.656 (0.27)	0.657 (0.27)	0.657 (0.27)	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)	0.651 (0.22)
DRUP	0.656 (0.27)	0.657 (0.27)	0.657 (0.27)	0.656 (0.27)	0.656 (0.27)	0.656 (0.27)	0.651 (0.22)

Table S11: Best nDCG over each operator’s hyper-parameter grid under a cap on the Gini coefficient of top- K exposure.

Data	Operator	Gini ≤ 0.3	≤ 0.4	≤ 0.5	≤ 0.6	no cap
Coat	logged graph	–	0.492	0.492	0.492	0.492
	IPS adjacency	–	0.495	0.495	0.495	0.495
	DR adjacency	–	0.417	0.462	0.486	0.498
	DRUP	–	0.417	0.441	0.484	0.497
Yahoo! R3	logged graph	0.659	0.659	0.659	0.659	0.659
	IPS adjacency	0.657	0.657	0.657	0.657	0.657
	DR adjacency	0.657	0.657	0.657	0.657	0.657
	DRUP	0.657	0.657	0.657	0.657	0.657
KuaiRec	logged graph	–	–	–	–	0.603
	IPS adjacency	–	–	–	–	0.612
	DR adjacency	–	–	–	–	0.633
	DRUP	–	–	–	–	0.633

Table S12: Certificates under re-fitted nuisances: the nuisances are re-fitted on the poisoned log (no cross-fitting) and the realised top- K is checked against the frozen-nuisance certificates of the clean system.

Data	Operator	fake users F	hit rate (frozen / re-fitted)	certified pairs	violated
Coat	IPS	1, 2, 5, 10	0.001/0.001, 0.018/0.001, 0.268/0.001, 0.536/0.001	217	0
Coat	DR	1, 2, 5, 10	0.000/0.000, 0.000/0.000, 0.000/0.000, 0.004/0.000	5193	0
Coat	DRUP	1, 2, 5, 10	0.000/0.000, 0.000/0.000, 0.000/0.000, 0.004/0.000	5196	0
Yahoo! R3	IPS	1, 5, 20, 100	0.065/0.065, 0.065/0.065, 0.066/0.064, 0.174/0.064	0	0
Yahoo! R3	DR	1, 5, 20, 100	0.065/0.065, 0.065/0.065, 0.066/0.064, 0.174/0.064	0	0
Yahoo! R3	DRUP	1, 5, 20, 100	0.065/0.065, 0.065/0.065, 0.066/0.064, 0.174/0.064	0	0
KuaiRec	IPS	5, 50, 200	0.001/0.001, 0.001/0.001, 0.133/0.001	1616	0
KuaiRec	DR	5, 50, 200	0.000/0.000, 0.000/0.000, 0.500/0.000	2819	0
KuaiRec	DRUP	5, 50, 200	0.000/0.000, 0.000/0.000, 0.500/0.000	2821	0

Table S13: Jointly private DRUP for all fixed denoising ranks ($\delta = 10^{-5}$).

Data	rank	$\epsilon = 0.5$	1	2	4	8	16	∞
Coat	8	0.381	0.385	0.400	0.399	0.398	0.449	0.533
	32	0.382	0.385	0.403	0.385	0.401	0.438	0.548
	128	0.372	0.370	0.387	0.401	0.389	0.422	0.536
	full	0.376	0.389	0.378	0.393	0.394	0.417	0.532
	non-private DRUP (all nuisances) 0.555; popularity 0.527; 29 public users							
Yahoo! R3	8	0.483	0.492	0.491	0.493	0.492	0.493	0.492
	32	0.481	0.490	0.491	0.493	0.492	0.493	0.492
	128	0.479	0.490	0.490	0.494	0.493	0.492	0.493
	full	0.473	0.488	0.491	0.494	0.494	0.493	0.492
	non-private DRUP (all nuisances) 0.652; popularity 0.525; 1540 public users							
KuaiRec	8	0.175	0.179	0.173	0.427	0.603	0.618	0.634
	32	0.174	0.175	0.154	0.412	0.598	0.618	0.634
	128	0.160	0.173	0.172	0.365	0.581	0.614	0.634
	full	0.182	0.202	0.223	0.280	0.377	0.492	0.634
	non-private DRUP (all nuisances) 0.637; popularity 0.577; 717 public users							

Table S14: Semi-synthetic KuaiRec with propensities estimated by the logistic exposure model (columns as in Table 5; regret not computed).

p	Nuisances	C	Estimator	RMSE T	utility error (%)		regret	nDCG@20
					random	popular		
est.	independent log	fixed	IPS	77.7	+361.4±5.6	+138.2±1.1	–	0.057
			IPS+WC	12.2	+278.9±2.0	+107.8±0.9	–	0.057
			DR	92.2	+370.0±7.6	+146.3±1.1	–	0.029
			DRUP	21.2	+277.0±2.6	+107.2±0.9	–	0.038
est.	cross-fitted	fixed	IPS	78.4	+359.9±6.9	+136.1±1.0	–	0.056
			IPS+WC	12.0	+274.7±2.1	+105.8±0.7	–	0.056
			DR	113.8	+430.5±8.2	+155.3±1.1	–	0.021
			DRUP	27.2	+307.1±2.5	+114.9±0.8	–	0.030
est.	sample split	fixed	IPS	89.5	+536.5±6.4	+315.1±2.0	–	0.066
			IPS+WC	12.0	+453.5±2.6	+280.8±1.8	–	0.071
			DR	152.9	+544.2±8.7	+322.3±2.4	–	0.020
			DRUP	23.4	+446.6±3.0	+276.0±2.0	–	0.029
Five hops (independent log, true p , fixed C): relative bias of T_5 , DR 3.46, DRUP 0.32.								

Table S15: Semi-synthetic KuaiRec with a sparser log (density 3%, 58% of the propensities below the clip $\tau = 0.01$), true and estimated propensities (columns as in Table 5; regret not computed).

p	Nuisances	C	Estimator	RMSE T	utility error (%)		regret	nDCG@20
					random	popular		
true	independent log	fixed	IPS	326.6	+49.1±26.9	+23.2±1.1	0.816	0.083
			IPS+WC	11.4	-27.9±2.5	-12.1±0.5	0.816	0.083
			DR	341.1	+331.4±28.5	+121.2±1.7	0.839	0.037
			DRUP	39.9	+163.6±5.4	+64.2±1.2	0.841	0.046
true	independent log	$\hat{Y}$	IPS	—	-49.3±2.3	-42.4±0.4	—	0.086
			IPS+WC	—	-64.7±1.2	-54.2±0.2	—	0.086
			DR	—	-8.9±2.5	-30.7±0.4	—	0.159
			DRUP	—	-30.2±1.3	-42.9±0.3	—	0.144
true	cross-fitted	fixed	IPS	326.6	+49.1±26.9	+23.2±1.1	0.816	0.083
			IPS+WC	11.4	-27.9±2.5	-12.1±0.5	0.816	0.083
			DR	521.4	+484.6±30.7	+166.3±3.5	0.851	0.030
			DRUP	73.3	+260.3±8.1	+108.2±3.0	0.848	0.040
true	cross-fitted	$\hat{Y}$	IPS	—	-75.2±0.8	-57.9±0.1	—	0.098
			IPS+WC	—	-78.3±0.7	-61.4±0.1	—	0.098
			DR	—	-43.4±1.2	-46.6±0.1	—	0.140
			DRUP	—	-51.4±0.9	-52.1±0.1	—	0.133
true	sample split	fixed	IPS	291.2	+200.4±26.7	+197.0±5.6	0.855	0.052
			IPS+WC	9.0	+123.3±3.9	+158.9±5.3	0.855	0.049
			DR	3058.9	+1572.9±142.3	+848.2±69.1	0.865	0.020
			DRUP	294.4	+932.8±38.9	+583.1±29.9	0.863	0.029
true	sample split	$\hat{Y}$	IPS	—	-10.1±5.3	-19.5±1.4	—	0.156
			IPS+WC	—	-43.8±1.6	-39.6±0.7	—	0.141
			DR	—	+24.8±5.1	-8.5±1.4	—	0.121
			DRUP	—	-13.3±1.6	-28.7±0.8	—	0.132
est.	independent log	fixed	IPS	930.7	+333.6±32.8	+122.0±1.7	—	0.073
			IPS+WC	39.3	+121.1±5.0	+51.8±1.0	—	0.073
			DR	957.3	+891.6±39.0	+268.3±2.3	—	0.033
			DRUP	95.0	+516.4±10.6	+171.7±1.8	—	0.040
est.	cross-fitted	fixed	IPS	798.1	+331.0±33.1	+119.1±1.7	—	0.073
			IPS+WC	37.4	+112.7±4.7	+48.4±0.9	—	0.073
			DR	1074.9	+1242.2±48.4	+322.0±4.6	—	0.026
			DRUP	148.0	+703.3±18.2	+222.2±4.1	—	0.036
est.	sample split	fixed	IPS	940.7	+753.3±31.7	+514.8±10.0	—	0.062
			IPS+WC	40.0	+502.0±8.2	+415.5±8.9	—	0.064
			DR	14661.2	+2972.1±290.9	+1342.7±83.3	—	0.019
			DRUP	1239.1	+1836.6±68.0	+964.5±35.5	—	0.028

Table S16: AUC of the candidate list on unbiased test data (mean $\pm$ std over splits; ties count one half). Same methods, tuning and splits as Table 6, which tuned on the metric named there.

Method	Coat AUC	Yahoo! R3 AUC	KuaiRand AUC	KuaiRec AUC
<i>Non-personalised</i>				
Popularity	0.644	0.655	0.581	0.765
Imputation only	0.654	0.652	0.546	0.800
<i>Trained, pointwise loss</i>				
MF	0.575	0.640	0.528	0.802
IPS-MF	0.570	0.643	–	0.789
DR-MF	0.620	0.645	0.549	0.801
DR-JL	0.584	0.656	0.549	0.797
MRDR	0.589	0.657	–	0.793
iALS	0.667	0.761	0.596	0.727
LightGCN	0.632	0.705	0.559	0.799
DR-LightGCN	0.658	0.675	–	0.781
<i>Trained, pairwise or sampled loss</i>				
MF (BPR)	0.614	0.751	0.603	0.782
PDA	0.612	0.758	–	0.781
MACR	0.639	0.729	0.575	0.762
LightGCN (BPR)	0.657	<u>0.771</u>	0.614	0.785
r-AdjNorm	0.658	0.775	0.617	0.778
NAVIP	0.653	0.771	<u>0.614</u>	0.796
SimGCL	0.646	0.736	0.578	0.765
<i>Training-free, logged graph</i>				
linear LightGCN / r-AdjNorm	0.648	0.759	0.599	0.783
EASE	0.614	0.745	0.569	0.623
GF-CF	0.638	0.756	0.603	0.789
BSPM	0.665	0.756	0.604	0.789
<i>Training-free, debiased graph</i>				
IPS adjacency (NAVIP-style)	0.641	0.756	0.600	0.786
IPS + walk correction	0.641	0.756	0.600	0.786
propagation over $\hat{Y}$ only (no residual)	0.656	0.663	–	0.807
EASE on DR graph	0.663	0.702	0.545	0.788
GF-CF on DR graph	0.664	0.739	0.595	0.808
BSPM on DR graph	<u>0.666</u>	0.737	0.601	0.808
DR adjacency	0.652	0.756	0.600	<u>0.809</u>
DRUP	0.653	0.756	0.600	0.809
DR adjacency, 5 hops	0.655	–	–	0.808
DRUP, 5 hops	0.649	–	–	0.809
<i>Training-free, sample split (Assumption 2 holds)</i>				
DR adjacency, sample split	0.651	0.710	0.592	0.806
DRUP-split	0.650	0.710	0.592	0.806

Table S17: Recall at the cut-off of Table 6 (@5 on Coat and Yahoo! R3, @10 on KuaiRand, @20 on KuaiRec). On Yahoo! R3 the lists have ten candidates, so Recall@10 is 1 for every method and Recall@5 is reported.

Method	Coat Recall@5	Yahoo! R3 Recall@5	KuaiRand Recall@10	KuaiRec Recall@20
<i>Non-personalised</i>				
Popularity	0.644	0.679	0.509	0.166
Imputation only	0.663	0.678	0.480	0.186
<i>Trained, pointwise loss</i>				
MF	0.579	0.670	0.469	0.189
IPS-MF	0.573	0.671	–	0.192
DR-MF	0.624	0.675	0.481	0.188
DR-JL	0.593	0.688	0.481	0.194
MRDR	0.592	0.689	–	0.193
iALS	0.681	0.790	0.531	0.183
LightGCN	0.638	0.731	0.494	0.193
DR-LightGCN	0.671	0.704	–	0.190
<i>Trained, pairwise or sampled loss</i>				
MF (BPR)	0.622	0.778	0.531	0.178
PDA	0.623	0.787	–	0.176
MACR	0.645	0.750	0.513	0.179
LightGCN (BPR)	0.659	0.799	<u>0.542</u>	0.180
r-AdjNorm	0.662	0.802	0.543	0.180
NAVIP	0.661	<u>0.800</u>	0.541	0.185
SimGCL	0.656	0.755	0.508	0.162
<i>Training-free, logged graph</i>				
linear LightGCN / r-AdjNorm	0.646	0.781	0.529	0.181
EASE	0.620	0.767	0.509	0.154
GF-CF	0.657	0.777	0.532	0.183
BSPM	0.675	0.779	0.531	0.183
<i>Training-free, debiased graph</i>				
IPS adjacency (NAVIP-style)	0.645	0.783	0.528	0.185
IPS + walk correction	0.645	0.783	0.528	0.185
propagation over $\hat{Y}$ only (no residual)	0.664	0.699	–	0.192
EASE on DR graph	0.668	0.720	0.487	0.180
GF-CF on DR graph	0.672	0.760	0.526	0.194
BSPM on DR graph	<u>0.677</u>	0.756	0.528	0.194
DR adjacency	0.654	0.783	0.528	0.194
DRUP	0.655	0.783	0.528	<u>0.194</u>
DR adjacency, 5 hops	0.662	–	–	0.194
DRUP, 5 hops	0.655	–	–	0.194
<i>Training-free, sample split (Assumption 2 holds)</i>				
DR adjacency, sample split	0.659	0.745	0.521	0.193
DRUP-split	0.658	0.744	0.521	0.193

Table S18: Which nuisance breaks the correction under cross-fitting. Relative absolute bias of the three-hop term (40 users, 60 items, true propensities, 1,500 replications) with the degree weights C fixed a priori (degrees of Y), taken from the imputation $\hat{Y}$, or from the edge estimates W ; nuisances from an independent log, cross-fitted, fitted in sample, or from the 20% sample split. Compare IPS and DR with and without the walk correction.

Nuisances	Degree weights C	IPS, uncorrected	IPS+WC	DR, uncorrected	DRUP
independent log	fixed a priori	1.10	0.05	0.81	0.04
	from $\hat{Y}$	2.20	0.13	1.95	0.11
	from W	5.31	0.33	3.58	0.22
cross-fitted	fixed a priori	1.10	0.05	0.89	0.05
	from $\hat{Y}$	0.39	0.64	0.61	0.80
	from W	3.22	0.82	3.95	0.74
in-sample	fixed a priori	1.10	0.05	0.51	0.23
	from $\hat{Y}$	0.43	0.71	0.59	0.70
	from W	0.94	0.97	0.80	0.87
sample split 20%	fixed a priori	0.89	0.07	0.58	0.07
	from $\hat{Y}$	1.10	0.21	0.92	0.21

Table S19: Key comparisons under three tests. Δ : mean per-user difference of the dataset metric (first method minus second). p_{Holm} : user-level paired t -test, Holm-adjusted over all comparisons of the dataset. p_{NB} : corrected resampled t -test over the split means (Nadeau and Bengio, 2003), Holm-adjusted over the same family. p_{TOST} : two one-sided user-level tests of equivalence within ± 0.005 nDCG, with the margins ± 0.002 and ± 0.01 in brackets as sensitivity; a small value means that the two methods are equivalent. “Identical rankings”: the two operators select the same configuration on every split, so no test applies. The margin is a convention, not derived from the data. Sample-split rows use the first draw.

Comparison (first minus second)	Coat	Yahoo! R3	KuaiRand	KuaiRec
DRUP vs DR adjacency	+0.0014; 1.000 / 1.000 / 0.034 [0.381, $< 10^{-3}$]	identical rankings	identical rankings	-0.0001; 1.000 / 1.000 / $< 10^{-3}$ [$< 10^{-3}$, $< 10^{-3}$]
DRUP vs DR adjacency, 5 hops	-0.0081; 1.000 / 1.000 / 0.809 [0.956, 0.299]	–	–	-0.0003; 0.170 / 1.000 / $< 10^{-3}$ [$< 10^{-3}$, $< 10^{-3}$]
DRUP-split vs DR-split	-0.0033; 1.000 / 1.000 / 0.295 [0.651, 0.019]	-0.0012; 1.000 / 1.000 / 0.001 [0.271, $< 10^{-3}$]	+0.0002; 1.000 / 1.000 / $< 10^{-3}$ [$< 10^{-3}$, $< 10^{-3}$]	-0.0001; 1.000 / 1.000 / $< 10^{-3}$ [$< 10^{-3}$, $< 10^{-3}$]
DRUP vs GF-CF on DR graph	-0.0143; 0.106 / 1.000 / 0.984 [0.998, 0.842]	+0.0247; $< 10^{-3}$ / 1.000 / 1.000 [1.000, 1.000]	+0.0025; 0.043 / 0.317 / $< 10^{-3}$ [0.716, $< 10^{-3}$]	+0.0004; 1.000 / 1.000 / $< 10^{-3}$ [$< 10^{-3}$, $< 10^{-3}$]
DRUP-split vs DRUP	+0.0087; 1.000 / 1.000 / 0.625 [0.717, 0.456]	-0.0059; $< 10^{-3}$ / 0.045 / 1.000 [1.000, 1.000]	-0.0092; $< 10^{-3}$ / 0.068 / 0.996 [1.000, 0.315]	-0.0035; 0.004 / 1.000 / 0.051 [0.949, $< 10^{-3}$]
DRUP vs DR-JL	+0.0658; $< 10^{-3}$ / 1.000 / 1.000 [1.000, 1.000]	+0.1350; $< 10^{-3}$ / 0.049 / 1.000 [1.000, 1.000]	+0.0739; $< 10^{-3}$ / $< 10^{-3}$ / 1.000 [1.000, 1.000]	+0.0044; 0.003 / 1.000 / 0.292 [0.983, $< 10^{-3}$]
DRUP vs MRDR	+0.0599; 0.003 / 1.000 / 1.000 [1.000, 1.000]	+0.1323; $< 10^{-3}$ / 0.024 / 1.000 [1.000, 1.000]	–	+0.0046; 0.003 / 0.339 / 0.367 [0.987, $< 10^{-3}$]
DRUP vs LightGCN (pointwise)	+0.0073; 1.000 / 1.000 / 0.585 [0.691, 0.398]	+0.0616; $< 10^{-3}$ / 0.441 / 1.000 [1.000, 1.000]	+0.0443; $< 10^{-3}$ / 0.003 / 1.000 [1.000, 1.000]	+0.0103; $< 10^{-3}$ / 0.030 / 1.000 [1.000, 0.567]
DRUP vs iALS	-0.0372; 1.000 / 1.000 / 0.968 [0.978, 0.941]	-0.0065; 1.000 / 1.000 / 0.643 [0.867, 0.190]	-0.0031; 1.000 / 1.000 / 0.119 [0.744, $< 10^{-3}$]	+0.0249; $< 10^{-3}$ / 0.069 / 1.000 [1.000, 1.000]
DRUP vs r-AdjNorm	-0.0290; 1.000 / 1.000 / 0.920 [0.943, 0.868]	-0.0187; $< 10^{-3}$ / 1.000 / 0.999 [1.000, 0.981]	-0.0101; $< 10^{-3}$ / 0.485 / 0.999 [1.000, 0.526]	+0.0296; $< 10^{-3}$ / 0.040 / 1.000 [1.000, 1.000]

Table S20: KuaiRec with the log thinned to a fraction q of its exposures (nuisances re-fitted, additive imputation, $\tau \in \{0.05, 0.01\}$, $\lambda \in \{0, 1\}$, selected on validation). nDCG@20 of the uncorrected and corrected operators, and the lowest top-20 overlap and Spearman correlation between the two rankings at matched configurations.

thinning q	density	DR	DRUP	DR 5 hops	DRUP 5 hops	top-20 overlap	Spearman
1	13.38%	0.6302	0.6302	0.6306	0.6306	0.971	0.999
0.1	1.34%	0.6097	0.6102	0.6109	0.6111	0.935	0.999
0.03	0.40%	0.6073	0.6069	0.6085	0.6081	0.985	1.000

Table S21: Clip τ against exposure invariance for DRUP under the same intervention, with frozen nuisances (η : mean $\pm$ std over draws).

Data	Operator, τ	nDCG	rank shift	η	logged pairs clipped
Coat	logged graph	0.4917	-0.0646	+1.034 $\pm$ 0.043	–
	DRUP, $\tau = 0.01$	0.4191	+0.0071	-0.021 $\pm$ 0.008	1%
	DRUP, $\tau = 0.02$	0.4357	+0.0187	-0.054 $\pm$ 0.013	6%
	DRUP, $\tau = 0.05$	0.4599	+0.0104	-0.033 $\pm$ 0.008	12%
	DRUP, $\tau = 0.1$	0.4805	+0.0068	-0.021 $\pm$ 0.004	25%
	DRUP, $\tau = 0.2$	0.4891	+0.0046	-0.009 $\pm$ 0.002	77%
Yahoo! R3	logged graph	0.6590	-0.1604	+1.000 $\pm$ 0.011	–
	DRUP, $\tau = 0.0005$	0.6136	-0.0477	+0.181 $\pm$ 0.113	0%
	DRUP, $\tau = 0.002$	0.6140	-0.0519	+0.195 $\pm$ 0.099	1%
	DRUP, $\tau = 0.01$	0.6239	-0.0878	+0.508 $\pm$ 0.055	15%
	DRUP, $\tau = 0.05$	0.6500	-0.1274	+0.825 $\pm$ 0.017	51%
	DRUP, $\tau = 0.1$	0.6570	-0.1394	+0.901 $\pm$ 0.013	67%
	DRUP, $\tau = 0.2$	0.6572	-0.1494	+0.959 $\pm$ 0.009	82%
KuaiRec	logged graph	0.6026	-0.2412	+0.997 $\pm$ 0.003	–
	DRUP, $\tau = 0.01$	0.5309	-0.0000	-0.002 $\pm$ 0.002	2%
	DRUP, $\tau = 0.05$	0.6353	+0.0033	-0.007 $\pm$ 0.001	7%
	DRUP, $\tau = 0.1$	0.6382	+0.0059	-0.012 $\pm$ 0.000	10%
	DRUP, $\tau = 0.2$	0.6359	+0.0099	-0.021 $\pm$ 0.000	16%

Table S22: Exposure concentration and user groups. PRU: popularity–rank correlation. ECB (KuaiRec, where item quality is known from the fully observed matrix): partial correlation between an item’s mean rank and its log exposure, controlling for its true like rate. User gap: nDCG of group 0 minus group 1 (men minus women on Coat, inactive minus active users elsewhere) with a bootstrap 95% interval.

Data	Operator	nDCG	PRU	ECB	Gini@K	user gap [95% CI]	coverage
Coat	logged graph	0.4917	0.394	–	0.340	+0.112 [+0.033, +0.190]	0.937
	IPS adjacency	0.4951	0.415	–	0.346	+0.108 [+0.027, +0.192]	0.937
	DR adjacency	0.4890	0.689	–	0.599	+0.111 [+0.024, +0.189]	0.667
	DRUP	0.4891	0.688	–	0.600	+0.111 [+0.025, +0.189]	0.663
Yahoo!R3	logged graph	0.6590	0.534	–	0.281	-0.046 [-0.071, -0.019]	1.000
	IPS adjacency	0.6572	0.515	–	0.272	-0.044 [-0.069, -0.018]	1.000
	DR adjacency	0.6572	0.515	–	0.272	-0.044 [-0.069, -0.018]	1.000
	DRUP	0.6572	0.515	–	0.272	-0.044 [-0.069, -0.018]	1.000
KuaiRand	logged graph	0.4475	0.776	–	0.430	+0.015 [+0.004, +0.026]	0.977
	IPS adjacency	0.4493	0.776	–	0.428	+0.014 [+0.003, +0.025]	0.975
	DR adjacency	0.4493	0.776	–	0.428	+0.014 [+0.003, +0.025]	0.975
	DRUP	0.4493	0.776	–	0.428	+0.014 [+0.003, +0.025]	0.975
KuaiRec	logged graph	0.6026	0.845	0.830	0.991	+0.008 [-0.012, +0.029]	0.152
	IPS adjacency	0.6123	0.875	0.801	0.992	+0.014 [-0.007, +0.035]	0.081
	DR adjacency	0.6385	0.802	0.329	0.994	+0.012 [-0.009, +0.031]	0.012
	DRUP	0.6382	0.803	0.331	0.994	+0.012 [-0.009, +0.031]	0.012